Demographic change and entrepreneurship: General equilibrium implications for economic dynamism

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Abstract

Rapid aging and population decline pose a fundamental threat to economic dynamism. Using South Korea as a case study, this paper investigates how these demographic shifts erode entrepreneurship. We identify a critical but often overlooked mechanism, where the primary driver of decline is the rising price of labor induced by severe scarcity. As the workforce contracts, real wages rise significantly, increasing the opportunity cost of entrepreneurship. This shift generates "entrepreneurial brain drain" by raising the capability threshold for market entry; capable individuals who would otherwise create productive firms move into wage employment. Our analysis shows that this wage-driven selection effect accounts for 91.7% of the decline in the entrepreneurship rate, far outweighing demographic composition effects. Consequently, the economy experiences a sharp contraction in the extensive margin of firms, ultimately reducing long-run per capita gross domestic product by 28.6%. However, welfare analysis reveals a contrasting outcome: the same wage increase that erodes aggregate dynamism improves individual welfare, as higher labor income more than compensates newborn cohorts for the burden of increased longevity. We conclude that aging economies risk a "missing generation" of firms, as rising opportunity costs create a structural barrier to entry, highlighting a fundamental tension between aggregate dynamism and individual well-being.

Keywords: Demographic change, Population aging, Entrepreneurship, Occupational choice, General equilibrium model.
JEL Codes: E20, J11, J24, L26, D58.

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1 Introduction

Across the developed world, demographic trends characterized by declining fertility and rising longevity are reshaping economic landscapes. These shifts raise concerns not only about slowing economic growth but also about the erosion of key drivers of economic dynamism, particularly entrepreneurship. South Korea (hereafter Korea) provides a compelling case study, given its rapid aging trajectory and steep projected population decline. Although our model is calibrated to Korea, the results offer broader insights for other advanced economies facing similar labor supply constraints.

The core economic intuition of our analysis is straightforward. As the workforce contracts, labor becomes scarce, naturally driving real wages higher. For potential entrepreneurs, the market wage represents the opportunity cost of starting a business. As wages rise, the outside option of wage employment becomes increasingly attractive, making it more difficult to justify forgoing this income to pursue entrepreneurship. Crucially, this incentive shift raises the capability threshold for entry. Capable individuals who would otherwise establish productive firms are "drained" into the wage sector, as only the most exceptional entrepreneurs can justify the rising opportunity cost. We refer to this phenomenon as the "entrepreneurial brain drain." Unlike a demand-side shock arising from a shrinking consumer base, this supply-side price mechanism acts as a powerful structural filter that discourages new firm creation.

To quantify this mechanism, we develop a quantitative overlapping generations (OLG) model with heterogeneous agents and endogenous occupational choices. The model is calibrated to match the benchmark Korean economy in 2020. We then simulate the long-run equilibrium under the projected 2070 population structure and longevity. Demographic change affects entrepreneurship through two main channels: a composition effect in which changes in the aggregate pool of potential entrepreneurs are caused by shifts in age structure and a general equilibrium (GE) effect in which indirect effects operate through factor prices. Our analysis confirms the dominance of this GE mechanism.

Population decline induces severe labor scarcity, which raises real wages by 9.9% and substantially increases the opportunity cost of entrepreneurship. Quantitatively, this effect accounts for 91.7% of the decline in the entrepreneurship rate and contributes to a 28.6% reduction in long-run per capita gross domestic product (GDP). Importantly, however, the welfare analysis reveals a fundamental tension: despite the contraction in aggregate dynamism, individual welfare improves under GE because higher wages more than compensate newborn cohorts for the burden of increased longevity. The resulting structural reallocation from small and medium-sized enterprises (SMEs) to the corporate sector is therefore individually welfare-improving even as it diminishes aggregate dynamism.

This study bridges a critical gap between two strands of economic literature. The first examines the macroeconomic consequences of demographic change. Foundational work using OLG models has explored its implications for capital accumulation, fiscal sustainability, and

social insurance financing (e.g., Auerbach and Kotlikoff, 1987; Lim, 2016), while more recent studies investigate aging's effects on equilibrium interest rates (Carvalho et al., 2016) and secular stagnation (Eggertsson et al., 2019). Other contributions have investigated links between aging and productivity, focusing on channels such as automation (Acemoglu and Restrepo, 2022), workforce age structure (Feyrer, 2007), and aggregate labor productivity (e.g., Aksoy et al., 2019; Maestas et al., 2023). However, these studies have primarily emphasized the intensive margin of productivity or innovation. In contrast, our study highlights the extensive margin, specifically how demographic shifts alter the occupational choice between workers and entrepreneurs.

The second strand of literature has examined the determinants of entrepreneurship. Classical theories emphasize managerial ability (Lucas, 1978), risk preferences (Kihlstrom and Laffont, 1979), and the dynamics of learning and selection in industry evolution (Jovanovic, 1982; Hopenhayn, 1992). Complementary work on human capital has stressed the importance of balanced skill sets and accumulated experience for entrepreneurial success (Lazear, 2005; Gennaioli et al., 2013). Subsequent quantitative models have incorporated financial constraints and collateral channels to explain firm entry and dynamics (e.g., Evans and Jovanovic, 1989; Hurst and Lusardi, 2004; Cagetti and De Nardi, 2006; Lim, 2018). Empirical studies have also documented life-cycle patterns in entrepreneurial activity (Lévesque and Minniti, 2006) and linked demographic factors to startup rates (Liang et al., 2018).

We further contribute to the growing literature connecting population trends to business dynamism. Recent studies have shown that slowing labor supply growth can depress firm entry and increase market concentration (Hopenhayn et al., 2022; Karahan et al., 2024; Peters and Walsh, Forthcoming). We extend this literature by identifying and quantifying the specific role of the GE wage channel in a small open economy. Importantly, we show that the rising opportunity cost of labor acts as a powerful selection mechanism that reduces the number of entrepreneurs, a channel distinct from the pure labor supply quantity effects emphasized in previous studies. Moreover, by computing consumption equivalent variations (CEVs), we demonstrate that the resulting structural shift improves individual welfare despite the decline in aggregate GDP—a finding with important normative implications for policy evaluation in aging economies.

The remainder of the paper is organized as follows. Section 2 discusses the empirical background. Section 3 presents the structural model. Section 4 describes the calibration strategy. Section 5 presents the results of the quantitative analysis and decomposition. Section 6 concludes.

2 Empirical background

2.1 Demographic changes in Korea: Scale reduction and structural transformation

Korea is undergoing a fundamental demographic transformation at a pace unparalleled globally. The projected changes between 2020 and 2070, based on *Statistics Korea's Population Projections for Korea: 2020-2070* (medium variant), can be summarized by two features: scale reduction and structural transformation.

First, Korea's total population is projected to decline by approximately 28.3%, from 51.84 million in 2020 to 37.18 million in 2070 (see Table 1). This decline implies a contraction in both aggregate supply and aggregate demand.

Table 1: Projections of key demographic indicators: 2020 vs. 2070

Indicator	2020	2070	Note
Total Population	51,836,239	37,181,774	-28.3%
Working-Age Population Share (15-64)	72.1%	46.0%	-26.1 pp
Core Working-Age Population Share (25-44)	28.3%	16.7%	-11.6 pp
Elderly Population Share (65+)	15.7%	47.5%	+31.8 pp
Old-Age Dependency Ratio (65+ / 15-64)	21.8%	103.3%	+4.7-fold

Note: Author's calculations based on *Population Projections for Korea: 2020-2070*, published by Statistics Korea.

Second, a more severe issue is structural transformation, namely rapid population aging (Fig. 1). The share of the elderly population (65 and older) is projected to increase by approximately 31.8 percentage points (pp), rising from 15.7% to 47.5%. Conversely, the share of the working-age population (15–64) will decline sharply from 72.1% to 46.0%.

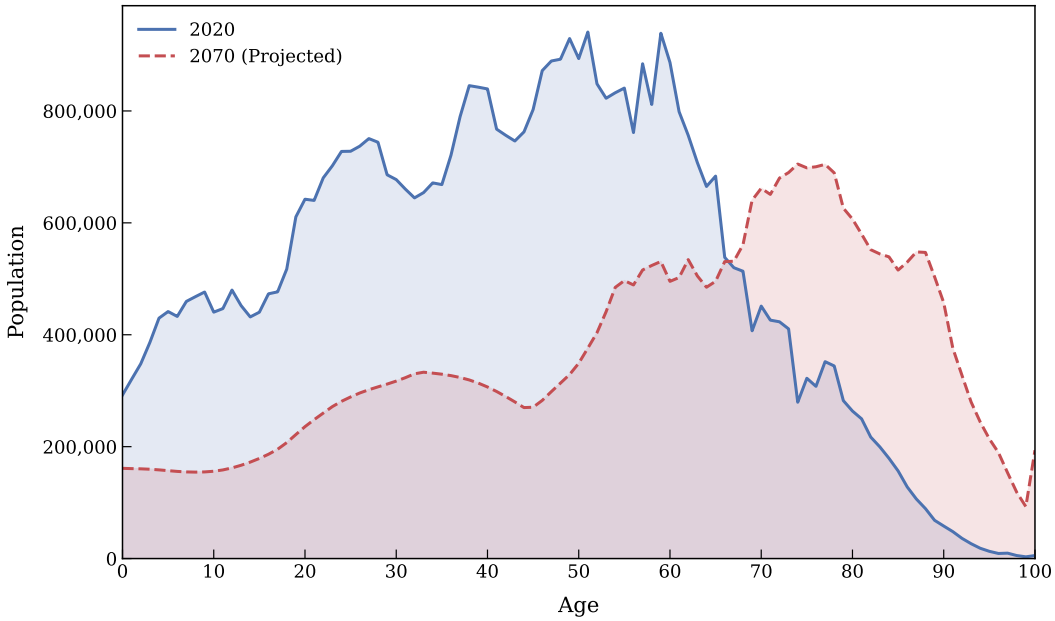
This structural shift dramatically increases the economic dependency burden. The old-age dependency ratio is projected to rise from 21.8% to 103.3%. By 2070, this implies that each working-age individual will support more than one elderly person on average, resulting in a shrinking labor supply and significant macroeconomic pressures.

2.2 Life-cycle entrepreneurship

Entrepreneurial activity is not uniformly distributed across the life cycle. Analysis of Statistics Korea's *Business Demography Statistics* (2019–2023 average) revealed a pronounced inverted U-shaped pattern in Korean entrepreneurship (Fig. 2).

The data showed a notably higher propensity to start a business during prime working ages (typically 30s–50s). The entry rate rises from 1.42% among individuals in their 20s to 3.11% in their 30s, peaking at 3.30% among those in their 40s. After this peak, the rate declines gradually to 2.89% in the 50s and 2.01% in the 60s. As a result, middle-aged cohorts dominate

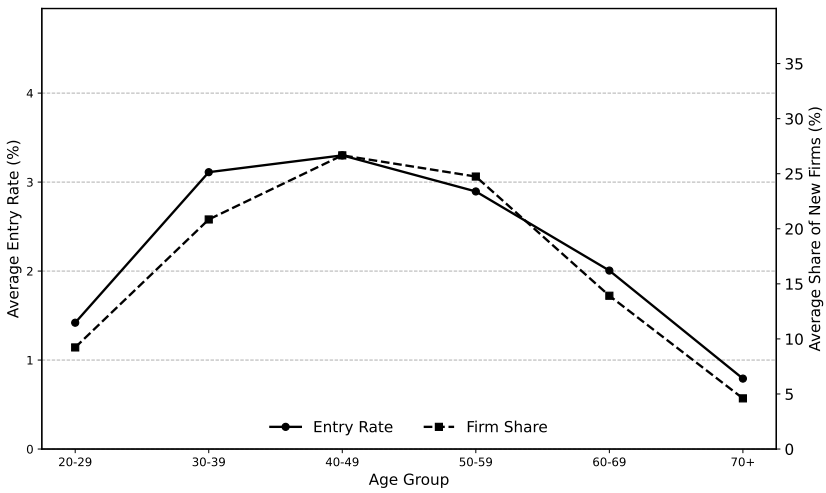
Figure 1: Population distribution by age: 2020 vs. 2070



Source: Population Projections for Korea: 2020-2070, published by Statistics Korea.

new firm creation. Individuals in their 40s account for the largest share of new firms (26.7%), followed closely by those in their 50s (24.7%) and 30s (20.8%). Combined, these three cohorts (30s–50s) account for approximately 72.2% of all new entrepreneurial activity.

Figure 2: Average entry rate and share of new firms by age group: 2019–2023 average



Source: Business Demography Statistics, published by Statistics Korea.

Theoretically, this life cycle pattern reflects dynamic changes in human capital accumulation, financing capacity, and risk preferences. The ongoing demographic transition in Korea, particularly the declining share of prime entrepreneurial age groups, may therefore influence aggregate entrepreneurial activity through two channels: the composition effect (a direct consequence of shifts in population structure and longevity) and the GE effect (indirect adjustments

through macroeconomic variables such as wages). We analyze and disentangle these two mechanisms.

3 The model

This study develops a structural model featuring OLG, heterogeneous agents, and endogenous occupational choice. The framework enables explicit modeling of demographic change, while endogenous occupational choice captures the dynamics of entrepreneurial activity and the SME sector.

3.1 Demographics

In the model, individuals enter the economy at age j_1, face an age-dependent mortality probability $\pi_d(j)$, retire at age j_R, and survive up to a maximum age J. Individuals younger than the retirement age j_R belong to the working-age population and make an occupational choice between becoming a worker or an entrepreneur (operating their own firm). Upon reaching retirement age j_R, individuals exit the labor force and can no longer engage in either occupation. Instead, they become eligible for a means-tested basic pension.

3.2 Households

Households derive utility from consumption c, with period utility given by $u(c) = \log(c)$. Future utility is discounted by β. Individuals are heterogeneous and characterized by four state variables: age (j), labor productivity (x), entrepreneurial ability (z), and assets (a).

The log of labor productivity, $\log x_t$, follows an AR(1) process:

$$\log x_t = \rho_x \log x_{t-1} + \epsilon_{x,t}, \quad \epsilon_{x,t} \sim N(0, \sigma_x^2), \tag{1}$$

where $\epsilon_{x,t}$ is an i.i.d. shock.

Entrepreneurial ability z is drawn from a log-normal distribution, $\log z \sim N(\mu_z, \sigma_z^2)$, when individuals enter the economy (age $j = j_1$). This ability is not permanent; it may expire at the end of each period with probability π_z. This parameter captures the exogenous probability of business termination due to unmodeled factors such as health shocks, skill obsolescence, or idiosyncratic market risks. If entrepreneurial ability is lost, the household draws a new value from the same distribution at the beginning of the next period. We assume that labor productivity x and entrepreneurial ability z are uncorrelated.[1]

[1] If they were positively correlated, high-ability individuals would command even higher market wages, reinforcing the opportunity cost mechanism emphasized in this study.

3.2.1 Working-age households

Economically active households $j < j_R$ face an occupational choice each period: worker or entrepreneur. The value function, $V^Y(j, x, z, a)$, is defined as the maximum of the worker value function, V^W, and the entrepreneur value function, V^E:

$$V^Y(j, x, z, a) = \max[V^W(j, x, z, a), V^E(j, x, z, a)]. \tag{2}$$

The occupational choice function, $O(j, x, z, a)$, is determined by this comparison:

$$O(j, x, z, a) = \begin{cases} 1 \text{ (Entrepreneur)} & \text{if } V^E(j, x, z, a) \geq V^W(j, x, z, a) \\ 0 \text{ (Worker)} & \text{if } V^E(j, x, z, a) < V^W(j, x, z, a) \end{cases} \tag{3}$$

Worker's Value Function. Workers choose consumption c and savings a' to maximize lifetime expected utility subject to a budget constraint:

$$V^W(j, x, z, a) = \max_{c,a'}\{u(c) + \beta(1 - \pi_d(j))E[V^Y(j + 1, x', z', a' \mid j, x, z)]\}, \tag{4}$$

subject to

$$(1 + \tau_c)c + a' = \{1 + (1 - \tau_k)r\}a + (1 - \tau_l)wxh(j). \tag{5}$$

The expectation term, $E[\cdot]$, is computed using the transition probabilities of x' and z'. The budget constraint describes resource allocation. The left-hand side includes consumption expenditures (incorporating the consumption tax, τ_c) and savings. The right-hand side includes after-tax capital income (interest rate r and capital income tax τ_k) and after-tax labor income. Labor income depends on the equilibrium wage, w, individual productivity, x, age-specific skill profile, $h(j)$, and labor income tax rate, τ_l. We assume households face a no-borrowing constraint, $a \geq 0$, implying no access to unsecured credit.

Entrepreneurial Value Function. Entrepreneurs also maximize lifetime expected utility. A key distinction is the inclusion of a nonpecuniary utility (or disutility) component from operating a business, $\chi_E(j)$, which may vary with age. This term captures age-specific preferences or institutional barriers that are not reflected in pecuniary returns.

$$V^E(j, x, z, a) = \max_{c,a'}\{u(c) + \chi_E(j) + \beta(1 - \pi_d(j))E[V^Y(j + 1, x', z', a' \mid j, x, z)]\}, \tag{6}$$

subject to

$$(1 + \tau_c)c + a' = \{1 + (1 - \tau_k)r\}a + (1 - \tau_f)\Pi(z, j). \tag{7}$$

The entrepreneur's income derives from business profits $\Pi(z, j)$, net of the business income tax rate, τ_f.

3.2.2 Retired households

Households retire at age j_R, permanently exiting both the labor market and the entrepreneurial sector. Retirees maximize their remaining lifetime utility by consuming their assets and receiving transfers.

$$V^O(j, a) = \max_{c,a'}\{u(c) + \beta(1 - \pi_d(j))V^O(j + 1, a')\}, \tag{8}$$

subject to

$$(1 + \tau_c)c + a' = \{1 + (1 - \tau_k)r\}a + b \cdot \mathbb{I}(a \leq \tilde{a}). \tag{9}$$

The transfer, b, represents a basic pension designed as a means-tested benefit that is provided only when assets fall below a threshold, $\tilde{a}$.[2]

3.3 Firms

The firm sector consists of two components: the entrepreneurial sector (SMEs) and the corporate sector (large firms).[3] Both sectors produce a homogeneous final good using capital and labor, but they differ in their underlying production technologies.

3.3.1 Entrepreneurial sector (SMEs)

This sector is operated by individual entrepreneurs and features decreasing returns to scale $\nu < 1$, consistent with the span-of-control theory (Lucas, 1978). The production function is

$$y^E = Z(z, j)(k^\alpha l^{1-\alpha})^\nu, \tag{10}$$

where $\alpha \in (0, 1)$ denotes the capital share. Total factor productivity (TFP) $Z(z, j)$ depends on the entrepreneur's ability, z, and age, j, capturing experience effects.

Entrepreneurs choose capital k and labor l as inputs to maximize profit:

$$\Pi(z, j) = \max_{k,l}\{y^E - wl - (r + \delta)k\}, \tag{11}$$

where δ is the depreciation rate.

[2]Korea's public pension system remains relatively immature, which necessitates a means-tested basic pension to support elderly individuals with limited income. This specification reflects the institutional features of Korea's Basic Pension, which targets the lower 70% of the elderly population to alleviate old-age poverty. Similar means-tested old-age safety nets exist in other OECD economies—such as Australia's Age Pension, the U.K.'s Pension Credit, and Canada's Guaranteed Income Supplement—as instruments to protect low-income seniors while maintaining fiscal sustainability.

[3]This dual structure reflects key institutional features of the Korean economy, in which a small number of large conglomerates (*Chaebols*) dominate exports and innovation, while a large number of SMEs operate in lower-productivity sectors with high entry and exit rates.

3.3.2 Corporate sector (large firms)

The corporate sector consists of a representative firm with constant returns to scale technology:

$$Y^C = Z^C (K^C)^\alpha (L^C)^{1-\alpha}, \tag{12}$$

where Z^C represents TFP, and K^C, L^C denote aggregate capital and labor inputs in this sector.

The representative firm maximizes profits under perfect competition:

$$\Pi^C = \max_{K^C, L^C} \{Y^C - wL^C - (r+\delta)K^C\}. \tag{13}$$

Both sectors are subject to the same business income tax, τ_f.

3.4 Government

The government maintains a balanced budget, financing government consumption G and pension transfers B through tax revenue.

$$G + B = T_L + T_K + T_C + T_F. \tag{14}$$

Here, T_L, T_K, T_C, and T_F represent total revenues from labor income, capital income, consumption, and business income taxes, respectively. Government consumption G is determined residually after pension expenditures B.

3.5 Competitive equilibrium

We define the equilibrium for a given period, t (e.g., 2020 or 2070). To analyze the economy under a given demographic structure, even when that structure is transitional rather than stationary, we adopt an approach widely used in the quantitative macroeconomic literature (e.g., Kitao, 2015). We decompose the aggregate household distribution, μ_t, into the age distribution, $\mu_{t,j}^{age}$ (i.e., the measure of individuals at age j) and conditional distributions $\Gamma_{t,j}(x, z, a)$ and $\Gamma_{t,j}(a)$ (i.e., the distribution of economic states conditional on age j) for working-age households and retirees, respectively. The age distribution, $\mu_{t,j}^{age}$, is taken directly from external demographic data for period t. We refer to the resulting equilibrium as a snapshot equilibrium.

A snapshot equilibrium for period t, given the age distribution, μ_t^{age}, and mortality rate, $\pi_{d,t}(j)$, consists of a set of prices, $\{w_t, r_t\}$; household policy functions; firm factor demands; government policies; and conditional distributions $\Gamma_{t,j}$, such that the following conditions hold:

1. **Household optimization:** Given prices, policies, and mortality rates $\pi_{d,t}(j)$, household policy functions solve the maximization problems defined in Section 3.2;

2. **Firm optimization:** Given prices, firms choose factor demands that solve the profit-maximization problems defined in Section 3.3;

3. **Government budget balance:** The government budget constraint holds, based on the aggregate distribution μ_t;

4. **Market clearing:**

   - Labor market: Total labor demand equals total labor supply (aggregated using μ_t);[4] and
   - Capital market: Under the Small Open Economy (SOE) assumption (detailed in Section 4.1), the domestic interest rate, r_t, is fixed at the world interest rate.[5]

5. **Consistent distributions:** The conditional distributions, $\Gamma_{t,j}(x, z, a)$, are stationary with respect to the economic state variables, (x, z, a), given the household policy functions and the stochastic processes for x and z.[6]

4 Calibration

The model is calibrated to the Korean economy in 2020 using the equilibrium framework defined in Section 3.5. This approach ensures methodological consistency between the baseline calibration and the 2070 simulation. We use the observed 2020 age distribution and mortality rates as inputs. The calibration follows a two-step procedure: first, externally determined parameters are set based on empirical evidence or standard values in the literature; second, internal parameters are jointly calibrated to match key macroeconomic moments.

4.1 Externally determined parameters

These parameters are specified before solving the model equilibrium.

Demographics and Preferences. The model period corresponds to 1 year.[7] Individuals enter the economy at $j_1 = 25$, retire at $j_R = 65$ (National Pension eligibility age), and live up to a

[4]Korea's labor market can be regarded as effectively closed. Foreign residents accounted for only about 4.9% of the total population in 2023 (Statistics Korea), and the inflow of foreign workers is tightly regulated through temporary schemes such as the Employment Permit System, which restrict long-term settlement. These institutional features limit international labor mobility, supporting our assumption that labor shortages are primarily absorbed through domestic wage adjustments.

[5]Aggregate domestic assets A_t must equal the sum of total domestic capital demand, K_t, and net foreign assets. $A_t = K_t + NFA_t$.

[6]This implies that $\Gamma_{t,j}$ evolves consistently across ages according to the agents' optimal decisions, but we do not require the age distribution $\mu_{t,j}^{age}$ to remain invariant over time.

[7]Given the high turnover rate of small businesses in Korea (where the 1-year survival rate is approximately 64% (Statistics Korea, 2023)), a yearly frequency is appropriate for capturing the rapid dynamics of firm entry and exit.

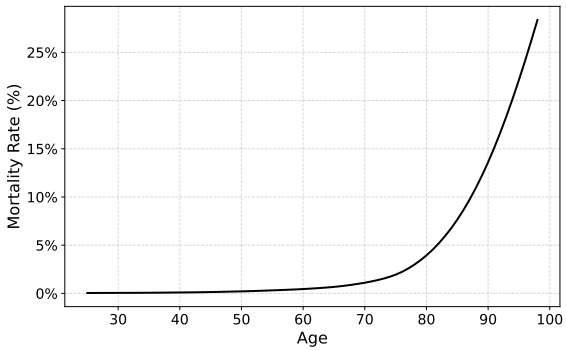
maximum age, $J = 99$. Age-specific mortality probabilities $\pi_d(j)$ are based on the *2020 Life Tables for Korea* published by Statistics Korea (see Fig. 3a).

Technology and Market Structure. We assume an SOE framework, fixing the real interest rate, r, at 3%, reflecting Korea's high degree of capital market integration.[8] Because global population aging may exert downward pressure on world interest rates, we also conduct a robustness check with a lower interest rate ($r = 2.5\%$) in Section 5.4. The capital income share, α, is set to 32.3% based on the National Accounts published by the Bank of Korea (BOK). The depreciation rate, δ, is 6.6% (Cho et al., 2012). The corporate-sector TFP, Z^C, is normalized to 1.0. The returns-to-scale parameter in the entrepreneurial sector, ν, is set to 0.88, consistent with standard values in the literature.

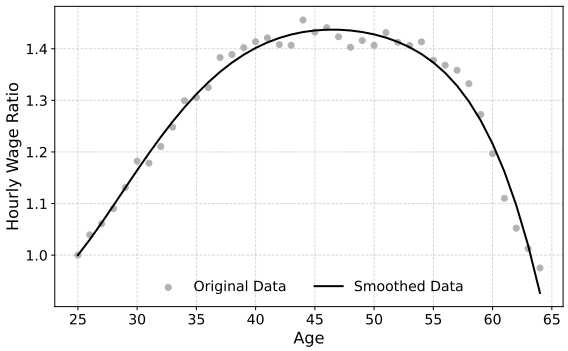
Stochastic Processes and Age Profiles. The persistence of labor productivity, ρ_x, is set to 0.92, following estimates for Korea by Chang et al. (2018). The expiration probability of entrepreneurial ability, π_z, is set to 10%.

Figure 3: Demographic and productivity profiles

(a) Age-specific mortality rate (2020)



(b) Age-dependent labor productivity



The age-dependent labor skill profile, $h(j)$, is estimated using wage data from the *Regional Employment Survey* conducted by Statistics Korea, consistent with the standard Mincerian wage equation (Mincer, 1974). Hourly wages are calculated by dividing average monthly earnings by average weekly working hours (scaled to a monthly basis) and are then smoothed using a fifth-order polynomial regression. The resulting profile is normalized by the wage level at age 25. As expected, the profile exhibits the typical inverted U-shape, peaking in the mid-40s (see Fig. 3b).

We also incorporate an age-dependent entrepreneurial experience effect, $g(j)$, into the SME production function, where $Z(z, j) = z \cdot g(j)$. This profile captures learning-by-doing effects

[8]Under the SOE assumption, excess domestic savings resulting from population aging accumulate as net foreign assets rather than depressing the domestic interest rate. Although this implies that domestic capital costs remain fixed, we address the potential effects of declining global interest rates through sensitivity analysis.

and the accumulation of human capital over the life cycle (Huggett et al., 2011). Specifically, we specify $g(j)$ using a concave function to capture the rapid accumulation of skills in the early stages of entrepreneurship that gradually stabilizes:

$$g(j) = \epsilon_g + (1 - \epsilon_g)(1 - e^{-\lambda_g(j-25)}), \tag{15}$$

where we set the base productivity, $\epsilon_g = 0.85$, and the learning rate, $\lambda_g = 0.15$. This parameterization implies that entrepreneurial productivity increases by approximately 15% over the life cycle. The functional form and calibration reflect the empirical regularity that surviving firms tend to grow as owners accumulate firm-specific managerial capital (Jovanovic, 1982).

Externally determined parameters are summarized in Table 2.

Table 2: Externally determined parameters

Description	Parameter	Value	Source/Basis
Demographics			
Entry Age	j_1	25	
Retirement Age	j_R	65	Pension Eligibility Age
Maximum Lifespan	J	99	Statistics Korea
Mortality Probability (2020)	$\pi_d(j)$	Age-specific	Life Tables for Korea
Technology and Preferences			
Real Interest Rate	r	3.0%	Small Open Economy
Capital Income Share	α	32.3%	National Accounts, BOK
Depreciation Rate	δ	6.6%	Cho et al. (2012)
Corporate Sector TFP	Z^C	1.0	Normalization
Returns to Scale (Entrepreneur)	ν	0.88	Standard Literature
Entrepreneurial Experience Base	ϵ_g	0.85	Learning-by-doing
Entrepreneurial Experience Rate	λ_g	0.15	Learning-by-doing
Stochastic Processes and Age Profiles			
Labor Productivity Persistence	ρ_x	0.92	Chang et al. (2018)
Entrepreneurial Ability Expiration Prob.	π_z	10.0%	Firm Exit Rate
Age-specific Labor Skill Profile	$h(j)$	Age-specific	Regional Employment Survey

4.2 Internally calibrated parameters

We jointly calibrate 11 key parameter sets $(\beta, \mu_z, \sigma_z, \sigma_x, \chi_E(j), \tau_l, \tau_k, \tau_c, \tau_f, \tilde{a}, b)$ endogenously to match corresponding moments of the 2020 Korean economy. Table 3 reports the calibrated parameters and their targets. The identification strategy follows the economic intuition linking each parameter to the corresponding empirical moment:

- **Preferences:** The subjective discount factor (β) governs household intertemporal savings decisions and is therefore a key determinant of aggregate capital accumulation. We calibrate β to match the total assets/GDP ratio of 3.9 reported in the *Korean National Balance Sheet*;

- **Entrepreneurial ability distribution:** The parameters governing the distribution of entrepreneurial ability (μ_z, σ_z) determine the scale and dispersion of the SME sector. The mean ability, μ_z, primarily affects the aggregate size of the SME sector; we therefore target the SME share of GDP (44.98%). The standard deviation, σ_z, governs firm heterogeneity, and we discipline this parameter using the Gini coefficient of firm output (0.812);

- **Labor productivity shocks:** The standard deviation of innovations to labor productivity, σ_x, determines the dispersion of wages among workers. We calibrate σ_x to match the standard deviation of log wages (0.546) observed in the *Regional Employment Survey* conducted by Statistics Korea;

- **Nonpecuniary utility of entrepreneurship:** The age-specific nonpecuniary (dis)utility, $\chi_E(j)$, introduces a structural wedge in the occupational choice decision. We adopt a two-step identification strategy to separate technological from preference effects. First, the technological productivity profile, $g(j)$, is fixed as described in Section 4.1. Second, taking $g(j)$ as given, we calibrate $\chi_E(j)$ for five age groups (20s, 30s, 40s, 50s, and 60s) as residuals. This approach enables the model to match the empirical age-specific entrepreneurship rates (the inverted U-shape), capturing age-dependent factors such as risk aversion or institutional barriers not explained by productivity alone; and

- **Fiscal policy:** The tax rates, τ_l, τ_k, τ_c, and τ_f, are calibrated to match effective tax revenue-to-GDP ratios, following the macro–public finance literature (e.g., Mendoza et al., 1994), since statutory tax rates often differ from the effective tax burden because of exemptions and deductions. The asset threshold for the basic pension, $\tilde{a}$, is calibrated to match the empirical receipt rate among the elderly (70%). The basic pension amount, b, is set to match the ratio of pension benefits to GDP per capita (8.43%).

Table 3: Internally calibrated parameters and targets

Parameter	Description	Target Moment (Value)
β	Subjective Discount Factor	Total Assets / GDP (3.9)
μ_z	Mean of Entrepreneurial Ability	SME Production Share (44.98%)
σ_z	Std. Dev. of Entrepreneurial Ability	Firm Output Gini Coefficient (0.812)
σ_x	Std. Dev. of Labor Productivity Innovation	Std. Dev. of Log Wages (0.546)
$\chi_E(j)$	Age-specific Nonpecuniary (Dis)utility	Age-specific Entry Rates
τ_l	Labor Tax Rates	Labor Tax Revenue / GDP (2.33%)
τ_k	Capital Tax Rates	Capital Tax Revenue / GDP (0.31%)
τ_c	Consumption Tax Rates	Consumption Tax Revenue / GDP (3.71%)
τ_f	Business Tax Rates	Business Tax Revenue / GDP (3.61%)
$\tilde{a}$	Asset Threshold for Basic Pension	Basic Pension Receipt Rate (70%)
b	Amount of Basic Pension	Pension Amount / GDP per Capita (8.43%)

Note: Data targets are based on 2020 Korean economy data.

The calibrated model replicates key features of the 2020 Korean economy and closely matches the life-cycle pattern of entrepreneurship (Table 4), accurately reproducing the inverted U-shape observed in the data. This fit supports the use of age-dependent nonpecuniary utility $\chi_E(j)$ as a residual wedge capturing age-specific factors influencing entrepreneurial entry.

Table 4: Entrepreneurship rate by age group: Data vs. model

Age Group	Model (%)	Data (%)
20s	1.42	1.42
30s	3.11	3.11
40s	3.27	3.30
50s	2.86	2.89
60s	2.00	2.01

Note: Data source is *Business Demography Statistics published by Statistics Korea (2019–2023 Average)*, corresponding to the rates detailed in Section 2.2.

5 Quantitative analysis

We conduct a comparative analysis to assess the long-term impact of demographic change by comparing the 2020 baseline snapshot equilibrium with the projected 2070 snapshot equilibrium.

5.1 Simulation design

To isolate the effect of demographic change, we hold the structural parameters calibrated to the 2020 economy (i.e., preferences, technology, and policies) fixed and introduce only the projected demographic changes for 2070. We implement the snapshot equilibrium approach, which relies on two key inputs:

1. **Mortality rates (for decisions):** We replace 2020 mortality rates with projected 2070 age-specific mortality probabilities $\pi_d^{2070}(j)$ directly in the household optimization problem. Specifically, agents use these projected survival probabilities when computing the expected continuation value in their Bellman equations, effectively adjusting the discount factor to $\beta(1 - \pi_d^{2070}(j))$. This ensures that individual decision rules (e.g., savings and occupational choices) endogenously incorporate the behavioral response to increased longevity.

2. **Population distribution (for aggregation):** We use the projected 2070 age distribution, μ_j^{2070}, as the demographic weights for aggregating individual policy functions into

economy-wide outcomes. This distribution is treated as exogenous and need not be stationary. Instead, it is imposed directly to compute aggregate factor supplies and demands. We scale the total population measure to reflect the projected change in the model-relevant population (ages 25–99), which is expected to decline by 18.4% between 2020 and 2070. Accordingly, if the 2020 population measure is normalized to 1.0, the 2070 measure is set to 0.816.

We then recompute the GE, allowing the wage rate, w, to adjust to clear the labor market under the new demographic conditions, while the interest rate, r, remains fixed under the SOE assumption.

5.2 Aggregate results

5.2.1 Macroeconomic aggregates and factor markets

The results reveal the substantial macroeconomic impact of the demographic transition (Table 5). Aggregate output Y is projected to decline by 41.8%. This contraction far exceeds the population decline (-18.4%), resulting in a sharp 28.6% reduction in per capita GDP. The magnitude of this decline indicates severe negative productivity effects beyond a simple scale effect.

Table 5: Key macroeconomic indicators: 2020 vs. 2070

Indicator	2020 GE	2070 GE	% Change
Total Population (25-99)	1.0000	0.8156	-18.4%
Aggregate Output, Y	0.8240	0.4796	-41.8%
Per Capita Gross Domestic Product (GDP)	0.8240	0.5880	-28.6%
Labor Supply	0.4220	0.2416	-42.8%
Real Wage	1.2792	1.4055	+9.9%
Assets/GDP	3.90	8.93	+129.0%

Significant GE effects arise from changes in factor scarcity. The shift toward an older population leads to a sharp 42.8% decline in aggregate labor supply. This pronounced labor scarcity raises the equilibrium real wage by 9.9%. Similarly, increased life expectancy (reflected in the projected 2070 mortality rates) strengthens saving incentives. As a result, the economy experiences substantial capital deepening, with the assets/GDP ratio rising by 129.0%.[9]

5.2.2 Entrepreneurship and industrial structure

The increase in wages strongly influences occupational choice, leading to a 26.2% decline in the overall entrepreneurship rate from 2.75% to 2.03%. This contraction is observed across all

[9]Under the SOE assumption, this excess saving is invested abroad as net foreign assets.

age cohorts (Table 6), indicating a broad-based erosion of entrepreneurial incentives over the life cycle.

Table 6: Changes in entrepreneurship rates across age cohorts: 2020 vs. 2070

Age Group	2020 GE (%)	2070 GE (%)	Change (pp)	Change (%)
Total	2.75	2.03	-0.72	-26.2
20s	1.42	1.02	-0.40	-28.2
30s	3.11	2.25	-0.86	-27.7
40s	3.27	2.61	-0.66	-20.2
50s	2.86	2.18	-0.68	-23.8
60s	2.00	1.37	-0.63	-31.5

This decline triggers a substantial industrial reorganization. SME sector output falls by 64.7%, far exceeding the 23.0% reduction observed in the corporate sector (Table 7). Consequently, the SME share of GDP contracts sharply from 45.1% to 27.4%, while the corporate sector's share rises to 72.6%, resulting in a highly concentrated and rigid industrial structure.

Table 7: Changes in sectoral output: 2020 vs. 2070

Sector	2020 GE	2070 GE	Change
SME Sector Output	0.3719	0.1314	-64.7%
(Share of GDP)	*(45.1%)*	*(27.4%)*	*(-17.7 pp)*
Corporate Sector Output	0.4522	0.3482	-23.0%
(Share of GDP)	*(54.9%)*	*(72.6%)*	*(+17.7 pp)*

5.3 Decomposing the decline in entrepreneurship: Composition vs. GE effects

To quantify the relative importance of the demographic composition effect (including changes in age structure and longevity) and the GE effect (the wage increase), we conduct a counterfactual simulation. Specifically, we compute a partial equilibrium (PE) snapshot for 2070 using the 2070 demographic structure (mortality rates and population distribution) while holding the real wage fixed at its 2020 level. The difference between the 2020 GE and 2070 PE isolates the composition effect, while that between the 2070 PE and the 2070 GE isolates the GE effect.

5.3.1 Decomposition results

Table 8 reports decomposition results. The PE simulation provides an important quantitative insight into the source of wage pressure. At the fixed 2020 wage, $w = 1.279$, the projected 2070 demographic structure generates a severe imbalance in the labor market: aggregate labor demand (0.3502) substantially exceeds labor supply (0.2402). This excess demand arises because

the working-age population contracts while firms' demand for labor, supported by high savings driven by longer life expectancy, remains relatively strong. The 9.9% wage increase observed in the 2070 GE therefore represents the price adjustment required to clear this imbalance.

Table 8: Decomposition of demographic change effects

	Equilibrium Outcomes			Decomposition of Changes		
Indicator	2020 GE (1)	2070 PE (2)	2070 GE (3)	Total Effect (3)-(1)	Composition Effect (2)-(1)	GE Effect (3)-(2)
Real Wage (level)	1.279	1.279	1.406	+9.9%	0.0%	+9.9%
Entrepreneurship						
Entrepreneurship Rate (%)	2.75	2.69	2.03	-0.72 pp	-0.06 pp	-0.66 pp
Relative Contribution (%)				*100.0%*	*8.3%*	*91.7%*
Macro Aggregates						
GDP (Y)	0.824	0.670	0.480	-41.8%	-18.7%	-28.4%
Per Capita GDP	0.824	0.821	0.588	-28.6%	-0.3%	-28.4%
Industrial Structure						
SME Share of GDP (%)	45.1	32.5	27.4	-17.7 pp	-12.6 pp	-5.1 pp
Corporate Share of GDP (%)	54.9	67.5	72.6	+17.7 pp	+12.6 pp	+5.1 pp

Note: 2070 PE assumes 2070 demographics (mortality and distribution) with factor prices fixed at 2020 levels. 2070 GE allows prices to adjust. The composition effect reflects the change due to demographics alone (2070 PE–2020 GE). The GE Effect reflects the change due to price adjustments (2070 GE–2070 PE). Changes in GDP and per capita GDP are multiplicative.

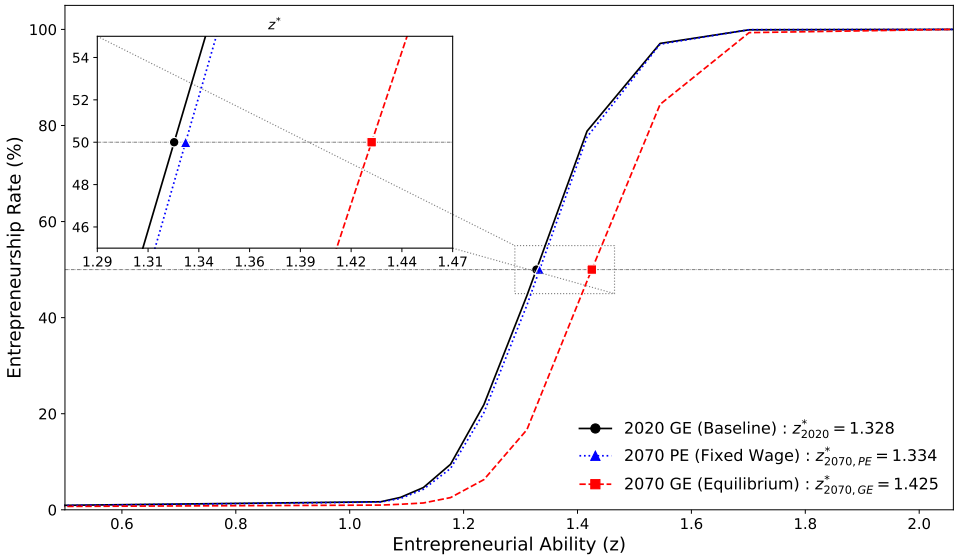
The decomposition yields a striking result: the GE effect overwhelmingly dominates the composition effect in explaining the decline in entrepreneurship. The total reduction in the entrepreneurship rate is 0.72 pp. Remarkably, only 0.06 pp (8.3%) is attributable to demographic composition (including longevity effects), while 0.66 pp (91.7%) arises from GE adjustments. This finding underscores that rising opportunity costs—driven by labor scarcity and wage pressure—constitute the primary channel through which demographic change erodes entrepreneurial dynamism.

5.3.2 Mechanism: Entrepreneurial brain drain

The predominance of the GE effect operates through an endogenous selection mechanism. Rising wages substantially increase the attractiveness of wage employment, thereby raising the critical ability threshold for entrepreneurship $z^\star$. As illustrated in Fig. 4, the 9.9% wage increase (GE effect) shifts the entry threshold from $z^\star \approx 1.334$ (in 2070 PE) to $z^\star \approx 1.425$.

This shift generates a phenomenon we term entrepreneurial brain drain. High-ability individuals ($z > z^\star_{old}$ but $< z^\star_{new}$) who would otherwise have established productive firms shift into wage employment.

Figure 4: Decomposition of the selection effect: Entrepreneurial brain drain



Note: The figure compares the entrepreneurship rate by ability z. $z^\star$ denotes the interpolated ability threshold where the rate exceeds 50% (magnified in the inset plot). The thresholds are $z^\star_{2020GE} \approx 1.328$, $z^\star_{2070PE} \approx 1.334$, and $z^\star_{2070GE} \approx 1.425$. The shift from 2020 GE to 2070 PE represents the Composition Effect, and the shift from 2070 PE to 2070 GE represents the GE effect (wage increase).

5.3.3 Implications for productivity and GDP

This selection mechanism, driven by rising wages, has significant implications for the firm productivity distribution. As shown in Table 9, the distribution becomes more compressed: the Gini coefficient of firm output declines from 0.773 to 0.734, and the production share of the top 10% of firms falls by 13.2 pp.

Table 9: Changes in firm productivity distribution statistics: 2020 vs. 2070

Indicator	2020 GE	2070 GE	Change
Firm Output Gini Coefficient	0.7727	0.7338	-5.04%
Bottom 90% Production Share	38.89%	52.11%	+13.2 pp
Top 10% Production Share	61.11%	47.89%	-13.2 pp

It is crucial to interpret these changes carefully. The increase in the entry threshold, $z^\star$, implies a strong selection effect: low-productivity firms are screened out of the market, raising the average productivity of surviving firms. From a GE perspective, however, this process represents a substantial contraction along the extensive margin. The flattening of the productivity distribution (a lower Gini coefficient) reflects the truncation of the left tail of the firm distribution, but more importantly, it indicates a loss of business diversity and a significant reduction in the total number of active firms.

This dominance of the extensive margin contraction explains the sharp decline in per capita GDP. Notably, Table 8 shows that under the composition effect alone (2070 PE), per capita GDP remains nearly unchanged (-0.3%). Although increased longevity induces substantial

capital deepening, reflected in the rise of the asset-to-output ratio (A/Y) from 3.9 in the 2020 GE to 5.933 in the 2070 PE scenario, this positive wealth effect is largely offset by the shrinking workforce.

However, once GE wage adjustments are introduced (2070 GE), the economy experiences a "missing generation" of firms. The wage increase reduces entrepreneurship and intensifies the entrepreneurial brain drain. The resulting loss of productive firms and reallocation of talent (from entrepreneurs to workers) generates a 28.4% decline in per capita GDP relative to the PE scenario. Consequently, the total decline in per capita GDP (-28.6%) is driven almost entirely by the GE effect. This result highlights the pivotal role of GE feedback in shaping macroeconomic outcomes during demographic transition.

5.3.4 Welfare implications of structural reallocation

The preceding analysis documented a substantial contraction of the SME sector alongside a sharp decline in aggregate GDP. However, because the corporate sector operates under constant returns to scale while the SME sector exhibits decreasing returns $\nu < 1$, the observed structural shift may partly reflect a reallocation toward a more productive sector rather than purely a loss of economic dynamism. To clarify the normative implications of this structural transformation, we compute a consumption equivalent variation (CEV) measure for newborn cohorts.

We evaluate the expected lifetime utility of a cohort entering the economy at age $j_1 = 25$:

$$W = \sum_{x,z} V^Y(j_1, x, z, 0) \cdot \pi_x^{erg} \cdot \omega_z, \tag{16}$$

where π_x^{erg} denotes the ergodic distribution of labor productivity, and ω_z denotes the cross-sectional distribution of entrepreneurial ability. The CEV, λ, represents the uniform percentage change in lifetime consumption that would make a 2020 newborn indifferent to being born into the counterfactual economy:

$$\lambda = \exp\left(\frac{W_{2070} - W_{2020}}{D}\right) - 1, \quad D = \sum_{t=0}^{J-j_1} \beta^t \prod_{s=0}^{t-1} (1 - \pi_d(j_1+s)). \tag{17}$$

Table 10 reports the results. The comparison between the GE and PE scenarios is particularly informative. Under PE, where wages remain fixed at the 2020 level, an individual born in 2070 is worse off ($\lambda = -5.18\%$). This welfare loss arises because increased longevity lengthens the retirement period without a corresponding increase in labor income, forcing individuals to spread resources over a longer horizon. In contrast, when wages adjust in GE, an individual born in 2070 is better off ($\lambda = +3.03\%$). The 9.9% wage increase more than offsets the longer retirement burden, producing a net welfare gain of approximately 8.2 pp relative to the PE scenario.

This result reveals a fundamental tension between individual welfare and aggregate dy-

Table 10: Welfare comparison: Consumption equivalent variation (CEV)

Scenario	Expected Utility (W)	CEV, λ (%)
2020 Benchmark	-19.220	—
2070 General Equilibrium (GE) (Baseline)	-18.451	$+3.03$
2070 Partial Equilibrium (PE) (Fixed Wage)	-20.590	-5.18

Note: W denotes the expected lifetime utility of a newborn cohort at age 25, averaged over the ergodic distributions of x and z with zero initial assets. CEV (λ) is computed relative to the 2020 benchmark using the benchmark effective discount factor sum $D = 25.77$.

namism. From the perspective of a representative newborn, the GE wage increase is unambiguously beneficial: higher wages raise labor income and expand lifetime consumption possibilities. From an aggregate perspective, however, the same wage increase weakens entrepreneurial incentives, contracts the extensive margin of firms, and reduces total GDP by 41.8%. The structural reallocation from SMEs to the corporate sector is therefore welfare-improving for individuals, reflecting a shift toward a more productive sector, even as it diminishes aggregate economic dynamism in the presence of population contraction.

5.4 Sensitivity and robustness analysis

In this section, we examine the robustness of the results to key assumptions about future economic conditions, with particular attention to potential long-run changes in global interest rates and fiscal closure rules. Table 11 summarizes the main equilibrium outcomes under these alternative scenarios.

Scenario 1: Global Aging and Lower Interest Rates. Because population aging is a global phenomenon, the world interest rate may decline in the long run. We simulate a counterfactual scenario in which the real interest rate falls to $r = 2.5\%$. The results show that cheaper capital increases capital accumulation (K rises from 1.687 in the baseline to 1.805). However, this higher capital intensity raises the marginal product of labor, pushing the equilibrium real wage even higher to 1.4243 (+11.3% relative to 2020). Consequently, the opportunity cost of entrepreneurship increases further, causing the entrepreneurship rate to decline to 1.92% (compared with 2.03% in the baseline). This confirms that the labor-scarcity channel dominates the cheaper financing channel: even with lower borrowing costs, rising wages suppress firm creation. Regarding individual welfare, the CEV remains positive ($+2.42\%$) but smaller than the baseline ($+3.03\%$), as reduced capital income partly offsets the wage gains from labor scarcity.

Scenario 2: Fiscal Closure with Fixed Government Consumption. Our baseline assumes that government consumption G adjusts residually to satisfy the budget constraint. In this scenario, we fix G at its 2020 level (0.0721) to reflect rigid public spending needs and allow the labor income tax, τ_l, to adjust. To finance higher spending and aging-related fiscal pressures,

Table 11: Robustness checks: Alternative scenarios for 2070

Indicator	2020 Benchmark	2070 Baseline GE ($r = 3.0\%$)	2070 Robustness 1 ($r = 2.5\%$)	2070 Robustness 2 (Fixed G)
Prices & Policy				
Interest Rate (r)	3.00%	3.00%	2.50%	3.00%
Real Wage (w)	1.2792	1.4055	1.4243	1.4069
Wage Change (%)	-	*+9.9%*	*+11.3%*	*+10.0%*
Gov. Consumption (G)	0.0721	0.0414	0.0366	0.0721
Labor Tax Rev. (T_L)	0.0192	0.0121	0.0122	0.0484
Entrepreneurship				
Ent. Rate (%)	2.75%	2.03%	1.92%	2.15%
SME Share of GDP (%)	45.1%	27.4%	28.6%	27.6%
Macro Aggregates				
GDP (Y)	0.8240	0.4796	0.4906	0.4795
Total Capital (K)	2.6828	1.6874	1.8046	1.6872
Assets/GDP (A/Y)	3.90	8.93	7.50	7.75
Welfare				
CEV, λ (%)	**—**	$+3.03$	$+2.42$	-8.17

Note: Robustness 1 simulates a decline in the global interest rate to 2.5% due to global aging. Robustness 2 maintains the absolute level of government consumption, G, at its 2020 level, financing the deficit by adjusting the labor income tax, τ_l. CEV (λ) is the CEV for a newborn cohort at age 25, computed relative to the 2020 benchmark (see Section 5.3.4 for methodology).

the government must substantially increase labor tax revenue (T_L rises from 0.0121 to 0.0484). The higher tax burden reduces after-tax wage income, making entrepreneurship marginally more attractive. As a result, the entrepreneurship rate recovers slightly to 2.15%. However, this improvement occurs alongside increased distortions. GDP ($Y = 0.4795$) remains essentially unchanged relative to the baseline ($Y = 0.4796$), and the economy still experiences a structural decline in dynamism compared with 2020. This experiment illustrates that although tax policy can influence entry at the margin, the underlying demographic drag on the economy persists. The welfare implications are particularly stark: the CEV turns sharply negative (-8.17%), as the substantial increase in labor income taxes required to maintain government consumption erodes the after-tax wage gains that would otherwise benefit individuals. Combined with the baseline welfare result ($+3.03\%$), this comparison shows that demographic change itself need not reduce individual welfare, but policy responses, particularly large tax increases, can impose substantial welfare costs on future cohorts.

Interpretation of the Quantitative Decomposition. Our quantitative attribution (i.e., 91.7% of the decline in entrepreneurship to GE wage adjustments) is conditional on the maintained calibration strategy. In particular, the age-dependent nonpecuniary disutility of entrepreneurship, $\chi_E(j)$, is calibrated as a residual to match observed age-specific entrepreneurship rates in the 2020 base year (Table 4) and is held fixed across demographic scenarios. This approach isolates changes in factor prices as the primary adjustment margin, abstracting by construction from potential shifts in nonpecuniary factors that may accompany demographic change. While this identification strategy is standard in the quantitative macroeconomics literature and enables clean identification of the GE wage channel, the precise magnitude of the 91.7% attribution should be interpreted within this framework. Nevertheless, the qualitative dominance of the wage channel, whereby labor scarcity raises wages and increases the opportunity cost of entrepreneurship, is a robust economic mechanism whose importance is unlikely to be overturned by alternative specifications of $\chi_E(j)$.

Discussion on Conservative Assumptions. Our model relies on several simplifying assumptions that likely render the results a conservative lower bound for the decline in economic dynamism.

First, regarding cohort productivity effects $h(j)$, we assume the age–productivity profile remains fixed. If future cohorts become more productive (e.g., through improved education or technological change), aggregate labor productivity, and thus equilibrium wages, would rise even further than in the baseline. This would intensify the opportunity cost of entrepreneurship and amplify the brain drain.

Second, regarding financial frictions, we impose a strict no-borrowing constraint ($a \geq 0$), implying that business formation relies entirely on self-financing. Demographic aging generates substantial capital deepening (A/Y rises from 3.9 to 8.93). In a model with collateralized

borrowing or relaxed liquidity constraints, this accumulation of wealth would loosen credit limits and could stimulate entry. By abstracting from this potential credit-driven expansion, the baseline specification isolates the pure negative GE effect of labor scarcity, preventing the demographic drag from being understated relative to the wealth effect.

6 Conclusion and policy implications

This study examines the long-term economic consequences of Korea's projected demographic transition, characterized by rapid aging and population decline, by 2070 using a structural OLG model with endogenous occupational choice and a snapshot equilibrium approach. The analysis shows that population decline generates a severe qualitative deterioration in economic dynamism, primarily by weakening entrepreneurship through dominant GE effects.

The key findings are as follows. First, severe labor shortages raise real wages by 9.9%. Second, higher wages substantially increase the opportunity cost of entrepreneurship, leading to a 26.2% reduction in the share of entrepreneurs. Decomposition analysis shows that GE effects account for 91.7% of this decline, far exceeding the contribution of demographic composition effects (8.3%). Third, the SME sector contracts sharply, with its GDP share falling from 45.1% to 27.4%, resulting in a more concentrated industrial structure. Fourth, the rising wage threshold triggers an entrepreneurial brain drain, as capable individuals shift into wage employment, producing a substantial contraction along the extensive margin. Together, these forces generate a 28.6% decline in per capita GDP. Fifth, despite this aggregate contraction, welfare analysis indicates that individual welfare improves under GE (a CEV of $+3.03\%$): the wage increase more than compensates newborn cohorts for the burden of increased longevity, implying that the structural reallocation is welfare-improving for individuals. However, this welfare gain depends on the fiscal response. Under a fiscal closure rule with fixed government consumption, the welfare effect becomes sharply negative (-8.17%), as large labor tax increases erode after-tax wage gains.

These results reveal a nuanced picture. Wage pressure arising from labor scarcity is the central mechanism constraining aggregate economic dynamism, yet the same mechanism improves individual welfare through higher labor income. The key policy implication is that this welfare gain is fragile: tax increases required to finance aging-related fiscal burdens can fully offset it. Policy responses should therefore mitigate adverse GE effects while promoting productivity-enhancing strategies and avoiding excessive fiscal burdens on the working population.

Finally, the findings suggest caution regarding policy interventions. In a competitive framework, industrial concentration driven by factor scarcity is not inherently inefficient; indeed, the welfare analysis shows that the structural reallocation benefits individuals. However, the brain drain implies a loss of innovation potential. Policy should therefore avoid artificially sustaining firm counts. Instead, interventions should focus on removing distortions (e.g., regulatory barriers or financial market inefficiencies) that raise the opportunity cost of entrepreneurship,

ensuring that high-potential entrants are not crowded out by demographic headwinds.

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